Causal statistics for treatment models

5. Regression Discontinuity Designs

Ecole Normale Supérieure–PSL, L3 Economics and Cogmaster, 2025

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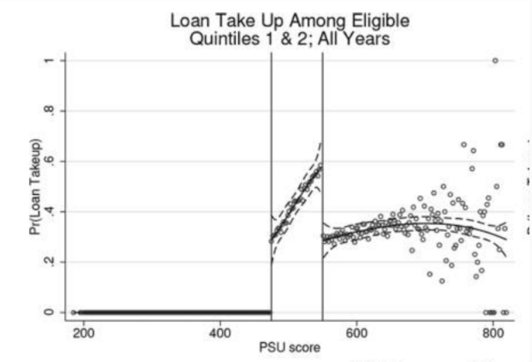
Today

We will detail the regression discontinuity strategy, relying carefully on the counterfactual representation. We will devote most time to the simple case of Sharp design, and see that intuition is simple, but estimation is complicated. We will then consider the more common case of Fuzzy design, and see that it simply brings us to the (now) familiar Wald and 2SLS estimation.

Starting example: Credit access and college enrolment

In Chile, public loans are available to pay for university fees, when a high school student scores above a threshold (475) on the national college admission test (PSU). Here is the proportion getting a loan as a function of the score (from low income families):



Nota: the second threeshold is for access to a bursary, forget it. Source: Solis, JPE 2017.

RDD argument

- People who are very close to 475 to the left

  and very close to 475 to the right

  are very similar, in the sense that + or - one point on a 600 scale is luck,

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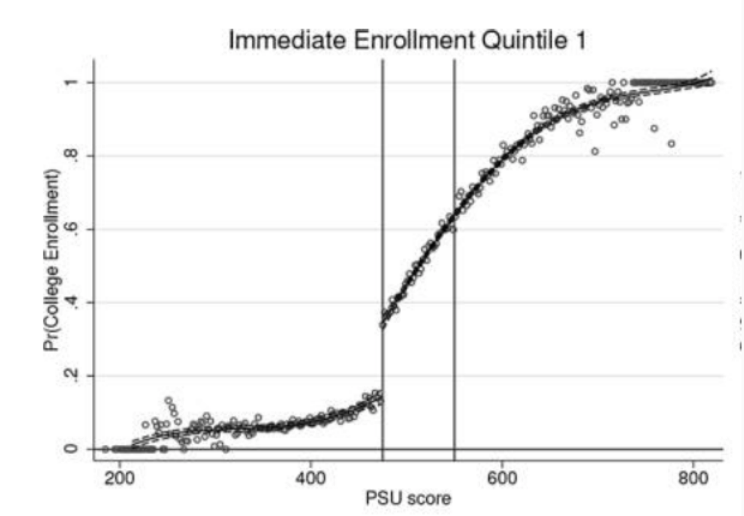
- **But**, there is one strong difference: people on the left don't have a loan

  whereas many people on the right do

- Thus comparing the outcomes of those two groups is similar to comparing two groups randomized into having a loan or not (with imperfect compliance, though, but we know we can deal with that)

Starting example cont'd

Question: Does accessing a loan increase college enrolment?



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Call S the **forcing variable** (in the above example, the PSU score)
and s_0 the threshold (in the above example 475)

We have a sharp design whenever:

$T = 1$ when $S \geq s_0$

and

$T = 0$ when $S < s_0$

Someone draws the graph of T as a function of S...

Counterfactuals

Let's specify the counterfactuals, **recognizing that they may be functions of the forcing variable**

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Start simple: assume linearity

$$
\begin{aligned}
Y(0) &= a + a'(S - s_0) + \varepsilon(0), \quad E(\varepsilon(0)|S) = 0 \\
Y(1) &= b + b'(S - s_0) + \varepsilon(1), \quad E(\varepsilon(1)|S) = 0
\end{aligned}
$$

- For ease of manipulation, we use $(S - s_0)$, but this is a normalization of the parameters
- $E(\varepsilon(k)|S) = 0$ seems more than a normalization; but under linearity of S, it is inconsequential (absorbed in a' or b')

“Counterfactuals may be functions of the forcing variable” is the most important thing to understand here

$\rightarrow$ interpret in the case of the loans in Chile

Estimand

The parameter of interest could be:

$$E(Y(1) - Y(0)|S) = [b - a] + [b' - a'](S - s_0)$$

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But, the form of this heterogeneity is sensitive to the linear functional form assumption. If counterfactuals where a quadratic function of S, we would have :

$$E(Y(1) - Y(0)|S) = [b - a] + [b' - a'](S - s_0) + [b'' - a''](S - s_0)^2$$

Estimand

A parameter that is not sensitive to functional form is:

$$E(Y(1) - Y(0)|S = s_0) = [b - a]$$

(This is the ATE for people who have S at the threshold value)

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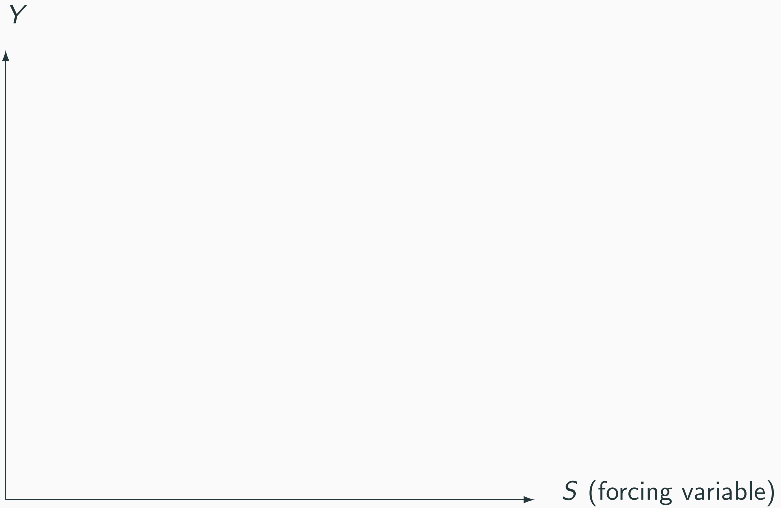
There is no quasi-experiment going on elsewhere, so causal effects at other points would need to rely on strong functional form hypothesis

Let's draw the two counterfctual functions



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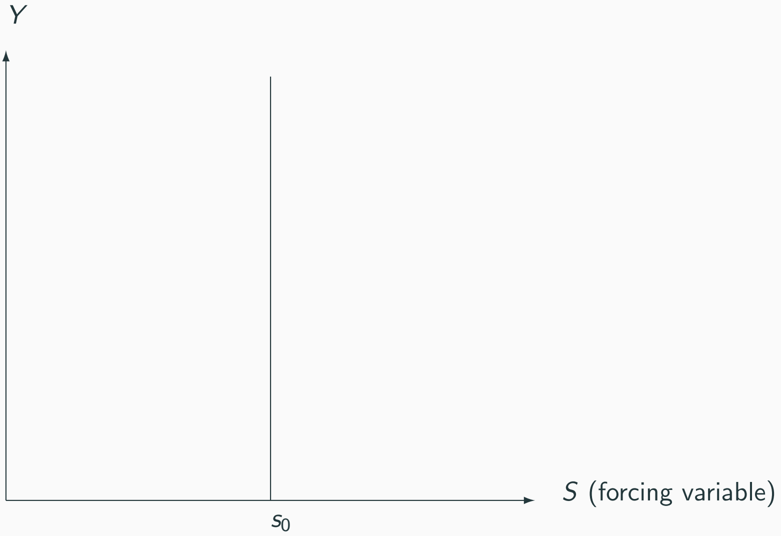

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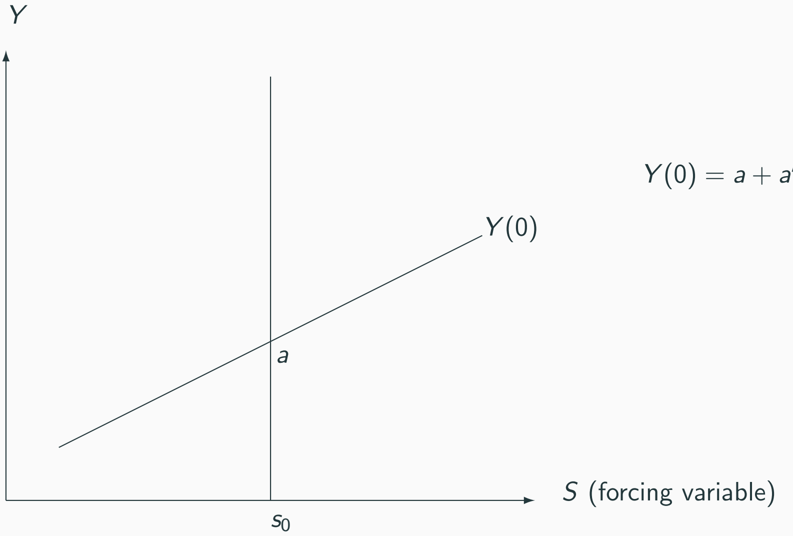

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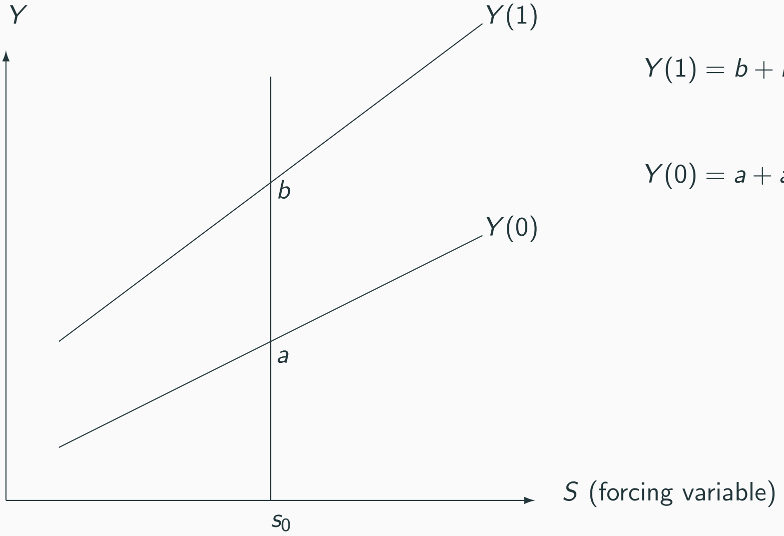
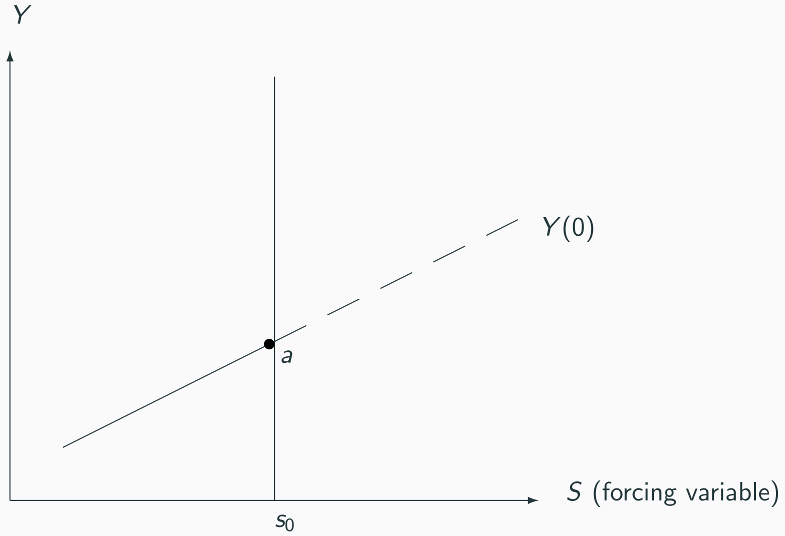


$$Y(1) = b + b'(S - s_0)$$

$$Y(0) = a + a'(S - s_0)$$

What we observe



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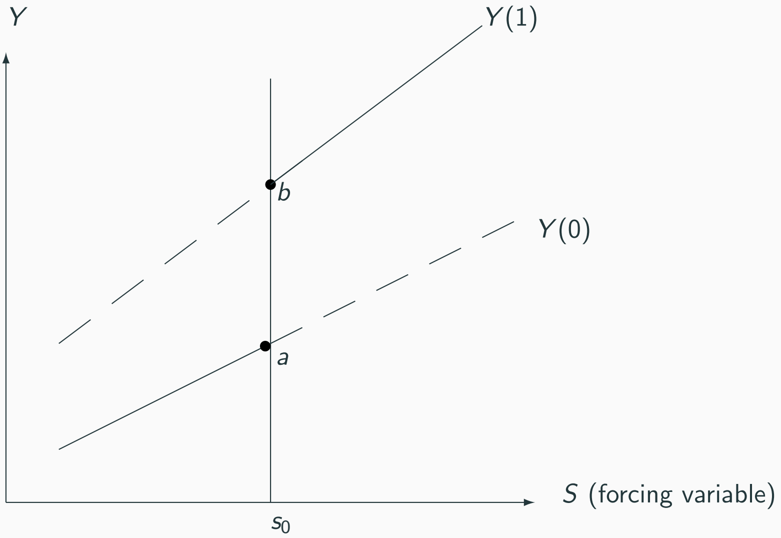
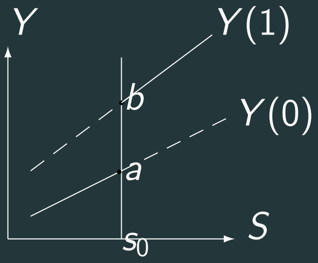


The estimation problem

Ideas to estimate $[b - a]$?



Simple averages ?

If we had a lot of data close to the threshold s_0, we could:

Estimate $E(Y(1)|S = s_0)$ as the empirical average of treated outcome at $S = s_0$ (point b)

and $E(Y(0)|S = s_0)$ would be approximated by the empirical average of untreated outcome *just below* $S = s_0$ (point a)

And the comparison of those two averages is meaningful because the two very close groups are super comparable

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And the comparison of those two averages is meaningful because the two very close groups are super comparable

But:

- We usually don't have that much data
- If the slope of $Y(0)$ is steep, relative to the effect $[b - a]$, the approximation of a could be rather bad

Simple averages ?

Rather, we could:

Run the regression $Y = b + b'(S - s_0) + \varepsilon(1)$ on the treated
If the linear specification is true, all the data points contribute to the estimation of b

Run the regression $Y = a + a'(S - s_0) + \varepsilon(0)$ on the untreated
If the linear specification is true, the exact value of a is obtained by extending the fitted curve to point s_0 (that is not in the untreated sample)

Equivalently

Use

$$Y = Y(0)(1-T) + Y(1)T$$

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\begin{aligned}
Y &= Y(0)(1-T) + Y(1)T \\
&= a + a'(S - s_0) + [b-a]T + [b'-a'](S-s_0) \times T + [\varepsilon(0)(1-T) + \varepsilon(1)T]
\end{aligned}
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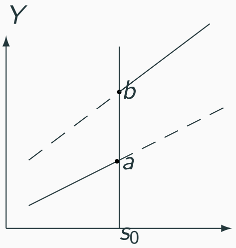
which fits the set of solid lines.



Equivalently

$$\begin{aligned} Y &= Y(0)(1-T) + Y(1)T \\ &= a + a'(S - s_0) + [b-a]T + [b'-a'](S-s_0)\times T + [\varepsilon(0)(1-T) + \varepsilon(1)T] \end{aligned}$$

As $T = \mathbf{1}(S \geq s_0)$ is only a function of S and the residuals are not functions of S,

T exogeneous and you can run this regression

But only true if the counterfactuals are really linear in S (otherwise S remains in the residual)

Identification

At this point, identification relies on two hypothesis:

1. The counterfactuals are linear in S
2. The counterfactuals are continuous functions of S (at least in a neighborhodd of s_0)

Start with 2. Think of what happens if true model is:

$$
\begin{aligned}
Y(0) &= a + a'(S - s_0) + \delta \mathbf{1}(S \geq s_0) + \varepsilon(0) \\
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$$Y = a + a'(S - s_0) + [b - a - \delta]T - \delta \mathbf{1}(S \geq s_0) + [b' - a'](S - s_0) \times T + \ldots$$

No way we can identify $[b - a - \delta]$ separately from δ because $\mathbf{1}(S \geq s_0)$ counfunded with T...

Identification

Any discontinuity must be driven exclusively by the shift from $Y(0)$ to $Y(1)$

In practice means that there must not be any other discountinous process that drives the counterfactuals at point s_0

For instance if 475 was also a threeshold for being admitted in a set of elite colleges...

More flexible forms

You don't have to trust that counterfactuals are linear in S. Assume they are quadratic:

$$\begin{aligned} Y(0) &= a + a'(S - s_0) + a''(S - s_0)^2 + \varepsilon(0), \quad E(\varepsilon(0)|S) = 0 \\ Y(1) &= b + b'(S - s_0) + b''(S - s_0)^2 + \varepsilon(1), \quad E(\varepsilon(1)|S) = 0 \end{aligned}$$

and

$$\begin{aligned} Y &= a + a'(S - s_0) + a''(S - s_0)^2 + \\ &\quad [b - a]T + ([b' - a'](S - s_0) + [b'' - a''](S - s_0)^2) \times T + \\ &\quad [\varepsilon(0)(1 - T) + \varepsilon(1)T] \end{aligned}$$

Exemple: grade retakers

In New York City public schools students pass a test: if they do not reach a certain performance on that test, they are retained in their grade for a year. Look at the impact of retention on highest grade achieved. This shows Y with threshold $s_0 = 0$. They use a quadratic fit.

Figure 7: Highest grade completed by summer assessment score

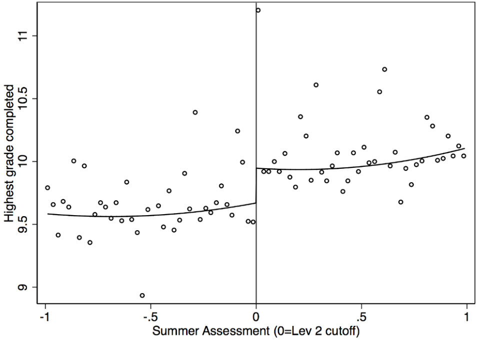


Figure depicts highest grade completed as a function of the summer assessment score. Circles represent cell means in bins of width 0.025. Solid line shows fitted values from a regression model of highest grade completed on a quadratic polynomial in the summer assessment score.

Source: Mariano et al. (2018)

Why stop there?

You could assume general functions of S, as long as they are **continous**!

$$
\begin{aligned}
Y(0) &= a + f(S) + \varepsilon(0), \quad E(\varepsilon(0)|S) = 0 \\
Y(1) &= b + g(S) + \varepsilon(1), \quad E(\varepsilon(1)|S) = 0
\end{aligned}
$$

and

$$
Y = a + f(S) + [b - a]T + [g(S) - f(S)] \times T + [\varepsilon(0)(1 - T) + \varepsilon(1)T]
$$

More flexible forms

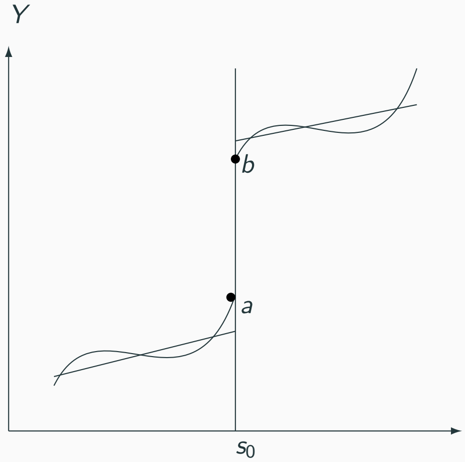
$$Y = a + f(S) + [b - a]T + [g(S) - f(S)] \times T + ...$$

We could estimate that with very flexible forms for f and g: this is the universe of non- (or semi-) parametric regression

Complicated and somewhat useless, because now, points away from $S = s_0$ do not help identify $E(Y(1) - Y(0)|S = s_0) = [b - a]$, which (remember) is the only causal parameter for which we have a quasi-experiment

More flexible forms

On another hand, if you restrict f and g somehow, you are forcing the arrival points a and b maybe very badly (here, for clarity, I force linear)



Local linear estimation

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At the end, there will always be *some* bias because you can only approximate the true f and g, and you usually do not have enough observations very close to s_0 to simply estimate averages (and what is "very close" anyway?)

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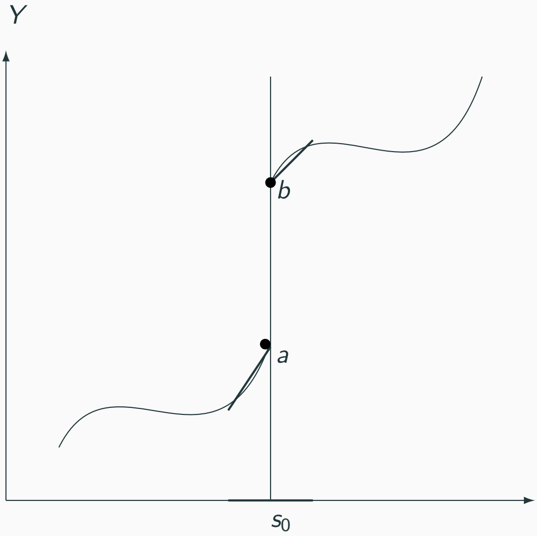
Let's do a bit of both: approximate the true shape of f and g **and** stay close to s_0:

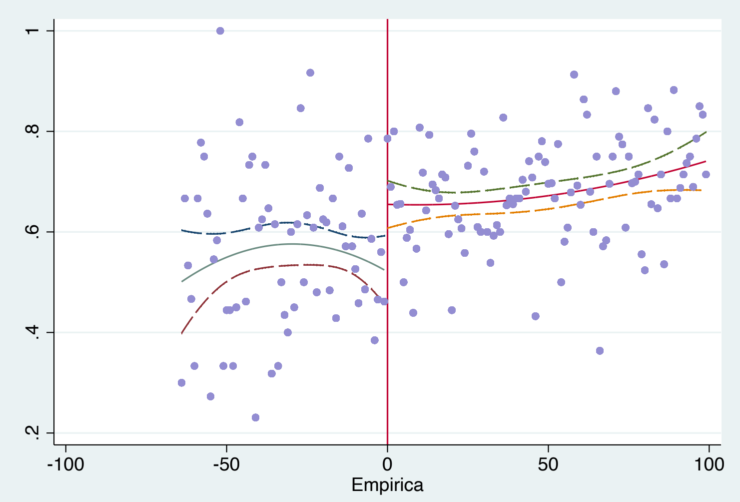
In a small neighborhood of s_0 (i.e. for values of S close to s_0), we can (first-order) approximate any function by a linear function

Local linear estimation



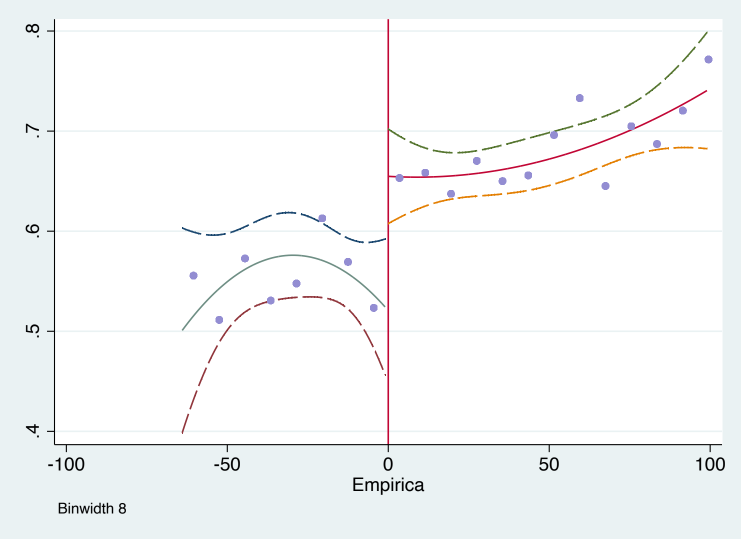
Loan and college, now in South Africa (Gurgand et al. 2023)

Full information:



Illustration

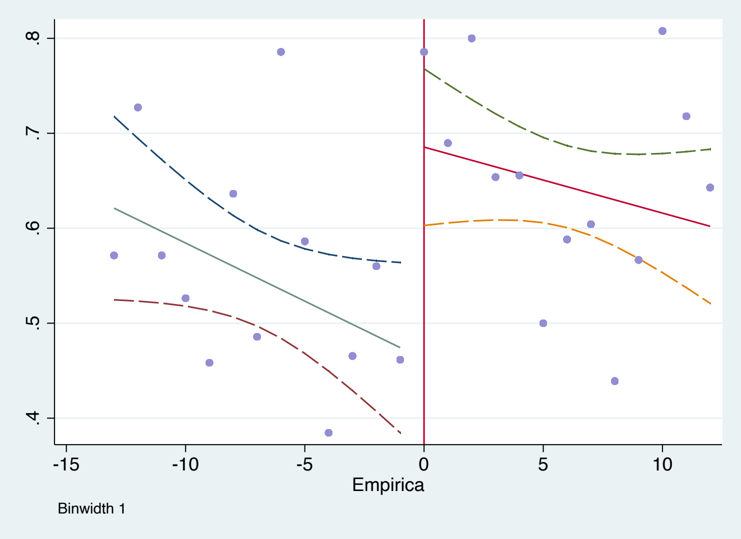
More visual (aggregate data):



$[b - a] = 0.13$ (p=0.002)

Illustration

Local linear estimation (neighborhood: +/- 13)



$[b - a] = 0.20$ (p=0.002)

Local linear estimation: arbitrage

Decisive question: choose the neighborhood around s_0

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- Too large: approximation is bad, $[b - a]$ has a large bias
- Too small: very little data, very imprecise estimator

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Balance the two using Mean Squared Error statistic:

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = Var(\hat{\theta}) + [Bias(\hat{\theta})]^2$$

Complicated methods to determine the estimation interval around s_0 that minimizes MSE (Calonico et al., 2014, Econometrica; Imbens and Kalyanaraman, 2012, Rev. Econ. Stud.)

Simple method for a complicated problem

After 27 slides, where do we end up?

1. Use one of the methods to determine the optimal estimation interval (at some point try to understand more precisely the logic of them)
2. Once you have it, restrict your data to this sample
3. Estimate the linear-approx. model by OSL:

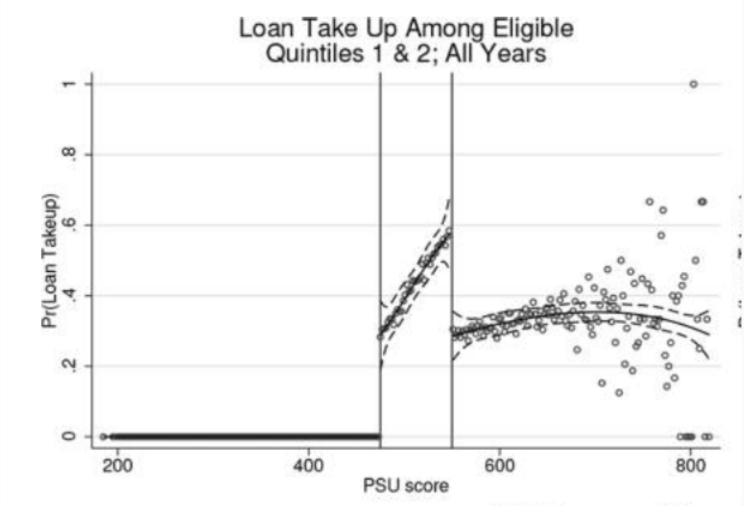
$$Y = a + a'(S - s_0) + [b - a]T + [b' - a'](S - s_0) \times T + ...$$

4. Always draw graphs (the "optimal" may not be very good if you don't have that much data)

Imperfect compliance again: the fuzzy design

Remind me: define formally the SHARP design

Let's look more carefully at the Chilean data: was that a sharp design?



Fuzzy design

We have a fuzzy design whenever:

$$P(T = 1|S \geq s_0) < 1$$

or

$$P(T = 0|S < s_0) < 1$$

(or both)

Extend ITT

Now, let's think in terms of Intention to treat:

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Z is just no longer identical to T

But you have an ITT concept:

- Just as much as you randomize Z but not all $Z = 1$ are actually treated,
- People may end-up (quasi-randomly) to the right of s_0 but not all will be actually treated

(and of course similarly for non-treated)

Extend ITT

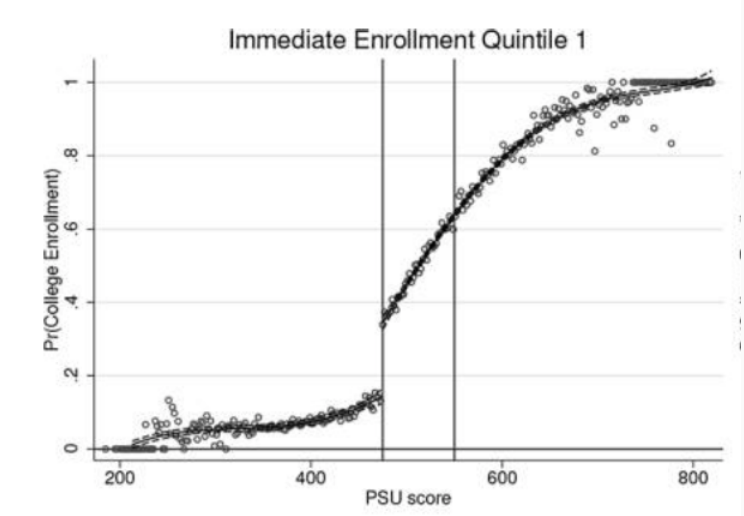
You can compare the outcome of people on the right vs. on the left by running the ITT regression:

$$Y = \alpha + \delta(S - s_0) + \beta Z + \gamma(S - s_0) \times Z + ...$$

on the optimal sample close to the threeshold (Everything we said earlier applies)

β is the ITT or reduced form parameter

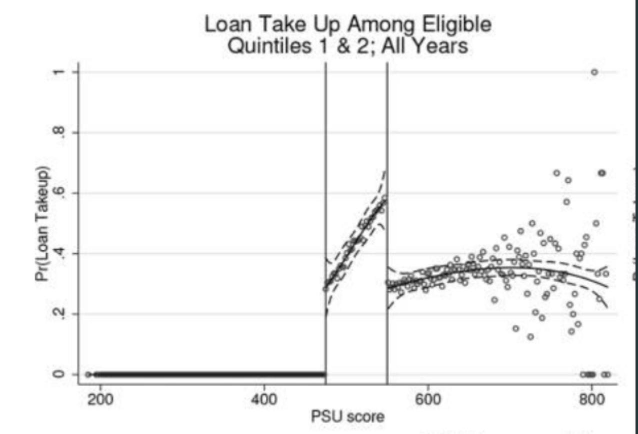
This is a reduced form



Extend the Wald estimator

Of course, the difference in outcomes measured by β must be scaled-down by the shares on the left and on the righ that do take the treatment, if you want to recover a treatment effect

How would you scale-down the ITT ? (assume ITT is 0.2)



FUZZY DESIGN = IMPERFECT COMPLIANCE

Everything we learned in the Imperfect compliance lecture applies:

- Distinction between T and Z
- Notion of ITT and of (differential take-up)
- Wald estimator
- 2SLS
- Compliers, always-takes, never-takers

The only trick is estimation, because we need to take into account the forcing variable

Estimation

There may be variants, but a simple way:

Think of Z as an instrument and T as the endogenous treatment

Estimation

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Think of Z as an instrument and T as the endogenous treatment

1. Preliminary: Graph the first-stage (what is that?): is there a clear jump, is it a fuzzy design indeed?
2. Consider the reduced form (ITT)
3. Run 2SLS

Consider the reduced form (ITT)

1. Find the optimal estimation interval for the reduced form
2. Keep the data on that interval
3. Run reduced form:

$$Y = \alpha + \alpha'(S - s_0) + \beta Z + \gamma(S - s_0) \times Z + ...$$

4. Significance and sign of β already tell you a lot
5. Graph the thing for transparency

Run 2SLS

1. Keeping the same optimal estimation interval:
2. First stage is:
$$T = \alpha' + \delta'(S - s_0) + \beta' Z + \gamma'(S - s_0) \times Z + ...$$
3. Second stage is:
$$Y = a + a'(S - s_0) + [b - a]\hat{T} + [b' - a']\widehat{(S - s_0) \times T} + ...$$
where $\hat{T}$ and $\widehat{(S - s_0) \times T}$ are instrumented in the first stage by Z and $(S - s_0) \times Z$

Note: last term in second stage could be $(S - s_0)Z$, it wouldn't matter (then one endo and one instrument: looks more familiar...)

Exercise for next week: run that 2SLS – we will talk about it

Today

We have explored the Regression discontinuity model. At the end of this lecture, you should:

- Know the distinction between a Sharp and a Fuzzy design
- Understand that the discontinuity provides a quasi-experiment, but only at the discontinuity threeshold
- Understand the graphical representation of the model using the counterfactual equations
- Understand the nature of the estimation problem and the local linear regression solution to it
- Understand the connexion between a Fuzzy design and imperfect compliance
- Know how to use the 2SLS estimation to estimate a Fuzzy design RDD model

Paper presentation: Nicolas Grimprel, “Mandatory Notice of Layoff and Job Search”